



Sampling Distributions & Hypothesis Testing

Lecture 4 - LAS

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Today

- Hypothesis Testing
- Sampling Distributions
- Central Limit Theorem
- Statistical Power



Today

[Chapter 7](#) and [Chapter 9](#) from the gitbook.



Learning Goals:

- What are hypotheses?
- What is the logic behind Null Hypothesis Significance Testing (NHST)?
- What is a sampling distribution?
- How do we test hypotheses?
- What is statistical power?



Hypothesis Testing



Hypotheses

What are hypotheses?

Examples:

- All swans are white
- A film director's gender has no impact on whether a movie passes the Bechdel test
- This medicine helps patients recover faster than no medicine





Falsification

- We typically cannot “confirm” or “validate” hypotheses.
- We can only reject them.
- If we observe 100 white swans, this does not tell us that all swans are white.
- If we observe one black swan, we can reject the hypothesis that all swans are white.





Falsification - Logic

1. Hypothesis/ Statement: All swans are white
2. Data: A black swan
3. Implication: Not all swans are white
4. Conclusion: Reject the hypothesis



Falsification - Requirements

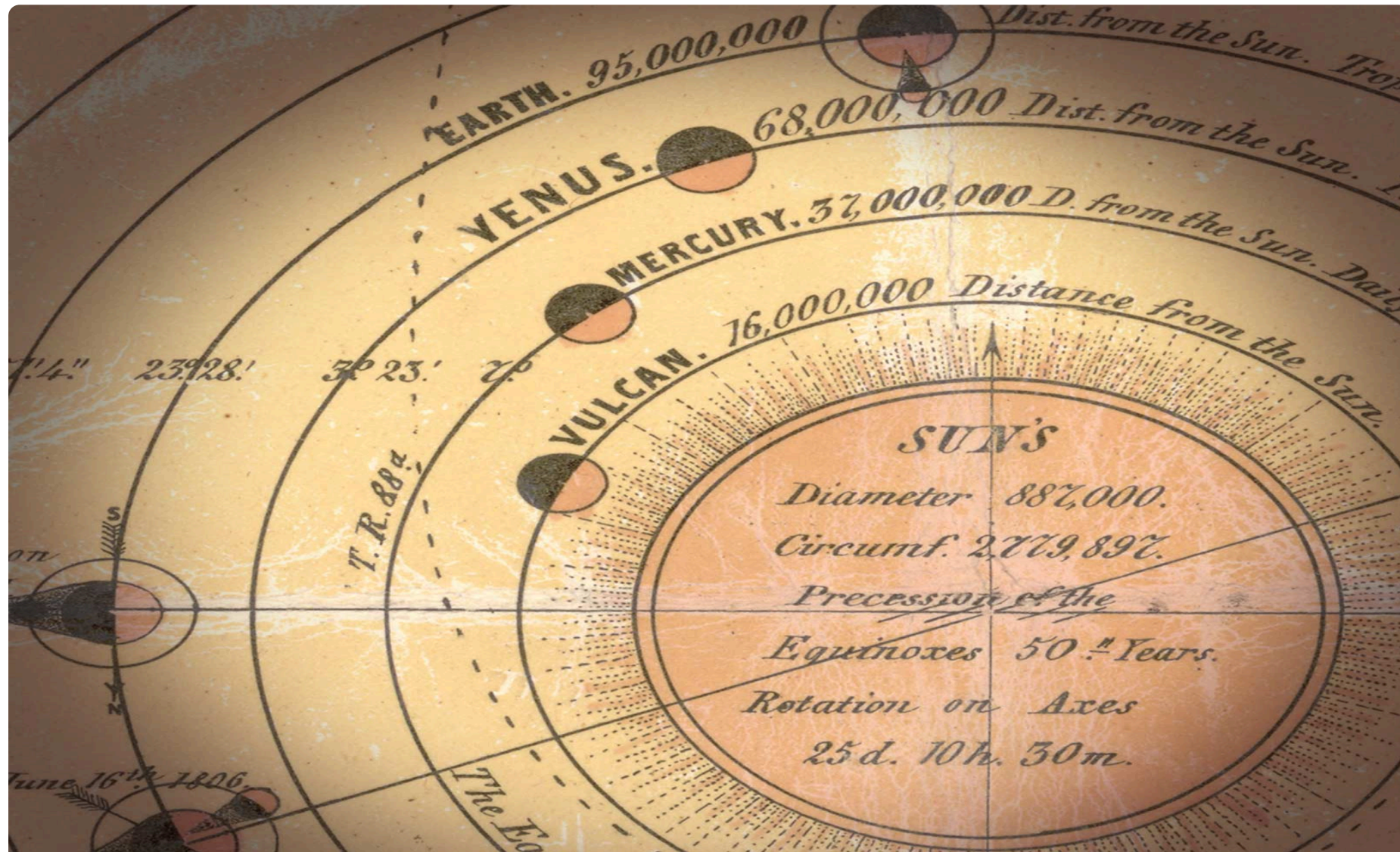
A hypothesis must be testable and capable of being proven false.

Is “God exists” a valid hypothesis?

No, because we cannot (scientifically) test the existence of god.

This has nothing to do with whether a claim is true or not. For it to be a hypothesis it must be *testable*.

Vulcan





From Hypotheses to Theories

Theories are the hypotheses that survive despite serious attempts to refute them.



Null Hypothesis Significance Testing

In our *sample* we observed that movies with male directors fail the Bechdel test more often than movies with female directors.

But what does this tell us about the population?

DIRECTOR GENDER				
Test	Female	Male	Mixed/other	Total row
FAIL	21	957	13	991
PASS	94	689	20	803
Total column	115	1646	33	1794



Null Hypothesis Significance Testing

We expect male and female directors to differ.

But we test the opposite:

- $\mathcal{H}_0$: male and female directors pass equally often.
- Assume $\mathcal{H}_0$ is true.
- Ask: how surprising are our observed data under $\mathcal{H}_0$?
- $p = P(\text{data at least this extreme} \mid \mathcal{H}_0)$
- If $p < \alpha$, reject $\mathcal{H}_0$.

This is *falsification under uncertainty*.



Falsification under Uncertainty

There are three options:

1. The data suggest to reject $\mathcal{H}_0$
2. The data are not informative to reject or accept $\mathcal{H}_0$
3. The data suggest to accept $\mathcal{H}_0$

NHST allows us to distinguish **1** from **not 1**:

- $p = P(\text{Data at least this extreme} | \mathcal{H}_0)$
- $p < \alpha \rightarrow \text{reject } \mathcal{H}_0.$
- $p \geq \alpha \rightarrow \text{do not reject } \mathcal{H}_0.$

Failing to reject $\mathcal{H}_0$ is not evidence for $\mathcal{H}_0$!



Incorrect Rejections: Error Rates

Even if $\mathcal{H}_0$ is true, unusual samples sometimes occur.

α controls how often we are willing to falsely reject $\mathcal{H}_0$.

- In the social sciences, usually $\alpha = .05$. When $\mathcal{H}_0$ is true, we falsely reject it in 1 out of 20 studies.
- Smaller α demands stronger evidence before rejection.
- For example, physics uses a 5 sigma rule, $\alpha = P(Z > 5)$
 $= 0.000000287$. A true $\mathcal{H}_0$ is falsely rejected in 1 out of 3.5 million studies.

This is a strength of NHST: Sampling uncertainty makes errors unavoidable, but we can control their long-run rate.



What a p -value Does *Not* Tell Us

The logic doesn't necessarily hold!

- $\mathcal{H}_0$: Piet is Dutch.
- Observation: Piet is a member of the Tweede Kamer (parlement).
- $P(\text{Tweede Kamerlid} \mid \text{Dutch})$ is extremely small.
- An NHST-like argument says: That observation would be extremely unlikely if Piet were Dutch, so reject $\mathcal{H}_0$.

Yet the observation is obviously evidence for Piet being Dutch rather than against it.

The problem: $P(\text{data} \mid \mathcal{H}_0) \neq P(\mathcal{H}_0 \mid \text{data})$

NHST computes the left-hand side.



NHST Summary

1. Define $\mathcal{H}_0$ and pick α .
2. Assume $\mathcal{H}_0$ is true.
3. Ask: how surprising are our observed data under $\mathcal{H}_0$?
4. $p = P(\text{data at least this extreme} \mid \mathcal{H}_0)$
5. If $p < \alpha$, reject $\mathcal{H}_0$.

In (3), NHST asks: “how surprising are our observed data if $\mathcal{H}_0$ were true?”

But how do we answer that question?



Sampling Distributions



Sampling Distributions

Recall the Bechdel-test dataset:

DIRECTOR GENDER			
Test	Female	Male	Total row
FAIL	21	957	978
PASS	94	689	783
Total column	115	1646	1761

We observed that female-directed movies fail less often than male-directed movies.

Could a difference this large arise simply because of *sampling variability*?



Sampling Distributions

Assume the null hypothesis is true:

- $\mathcal{H}_0$: Director gender and Bechdel-test outcome are unrelated.

Keep the totals fixed:

- 1761 movies
- 115 female-directed and 1646 male-directed
- 978 FAIL and 783 PASS

Under $\mathcal{H}_0$, the 978 FAILs occur randomly across the 1761 movies.

For every hypothetical repetition, record:

- How many of the 115 female-directed movies FAIL?



Sampling Distributions

If $\mathcal{H}_0$ is true, we expect the female-directed movies to have approximately the same failure rate as the complete sample.

The expected number of FAILs is therefore

$$\text{No. Female} \times \frac{\text{No. FAILs}}{\text{No. Movies}} = 115 \times \frac{978}{1761} \approx 63.9.$$

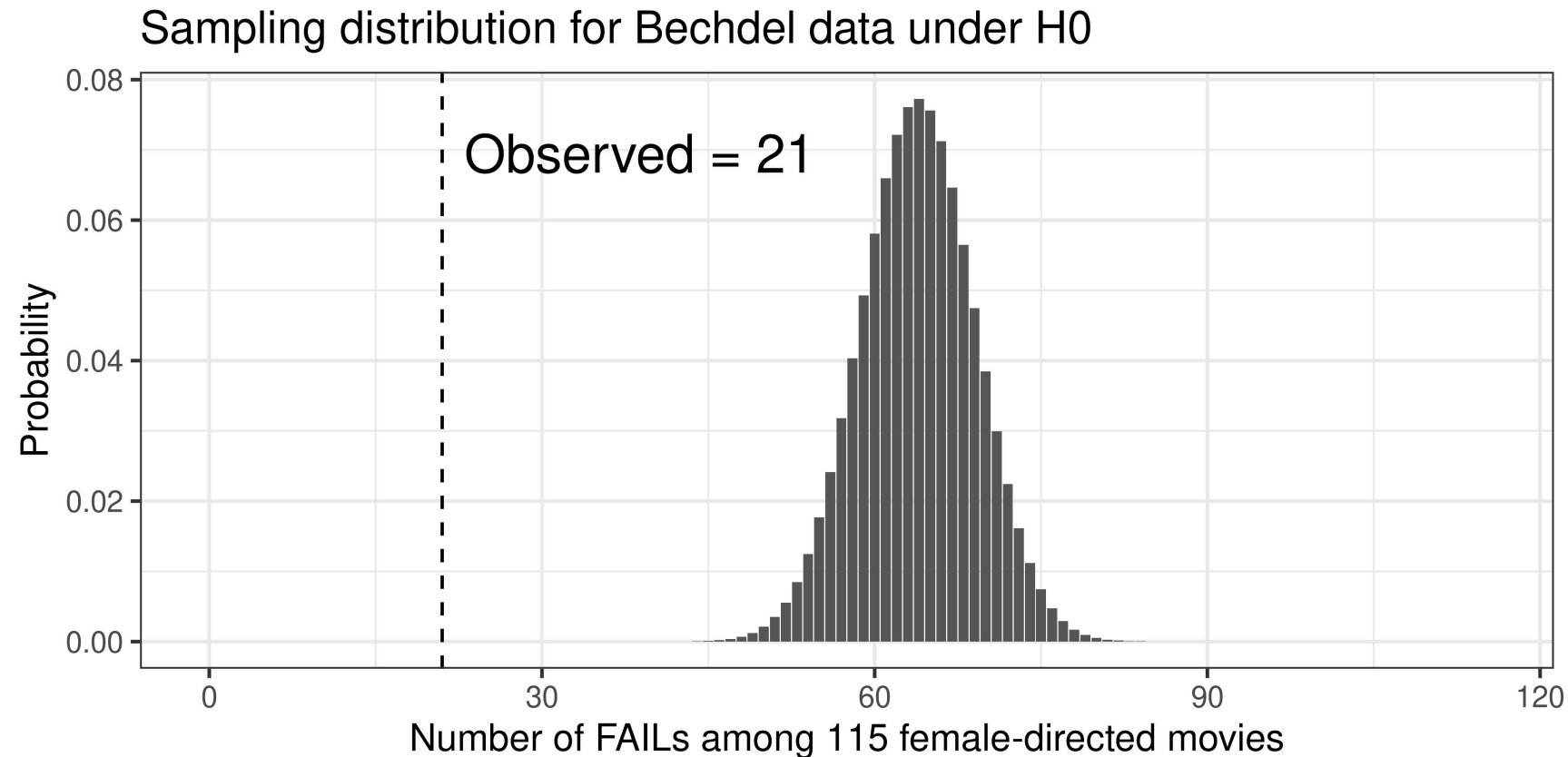
But because of sampling variability, we would not observe exactly 64 every time.

Sometimes we might observe 60, sometimes 70, sometimes 50, ...

The distribution of these hypothetical outcomes is a *sampling distribution*.



Sampling Distributions: Example



The observed no. FAILs lies far in the tail of the distribution.



From Sampling Distributions to p -values

The p -value asks:

Assuming $\mathcal{H}_0$ is true, how probable is a result at least as extreme as the one we observed?

Our observed value was $X_{\text{observed}} = 21$.

Under $\mathcal{H}_0$, observing *21 or fewer* FAILs among female-directed movies has probability $2.38 \times 10^{-17} \approx 0.\underbrace{000000000000000000}_{16 \text{ zeros}}238$

This is very unlikely, if $\mathcal{H}_0$ is true.



Alternative Hypothesis

So far we've only stated our null hypothesis.

- $\mathcal{H}_0$: Director gender and Bechdel-test outcome are unrelated.
- $\mathcal{H}_{1a}$: Director gender and Bechdel-test outcome are related.
 - $P(\text{PASS}|\text{Female}) \neq P(\text{PASS}|\text{Male})$
- $\mathcal{H}_{1b}$: Female directors pass the Bechdel-test more often than Male directors
 - $P(\text{PASS}|\text{Female}) > P(\text{PASS}|\text{Male})$

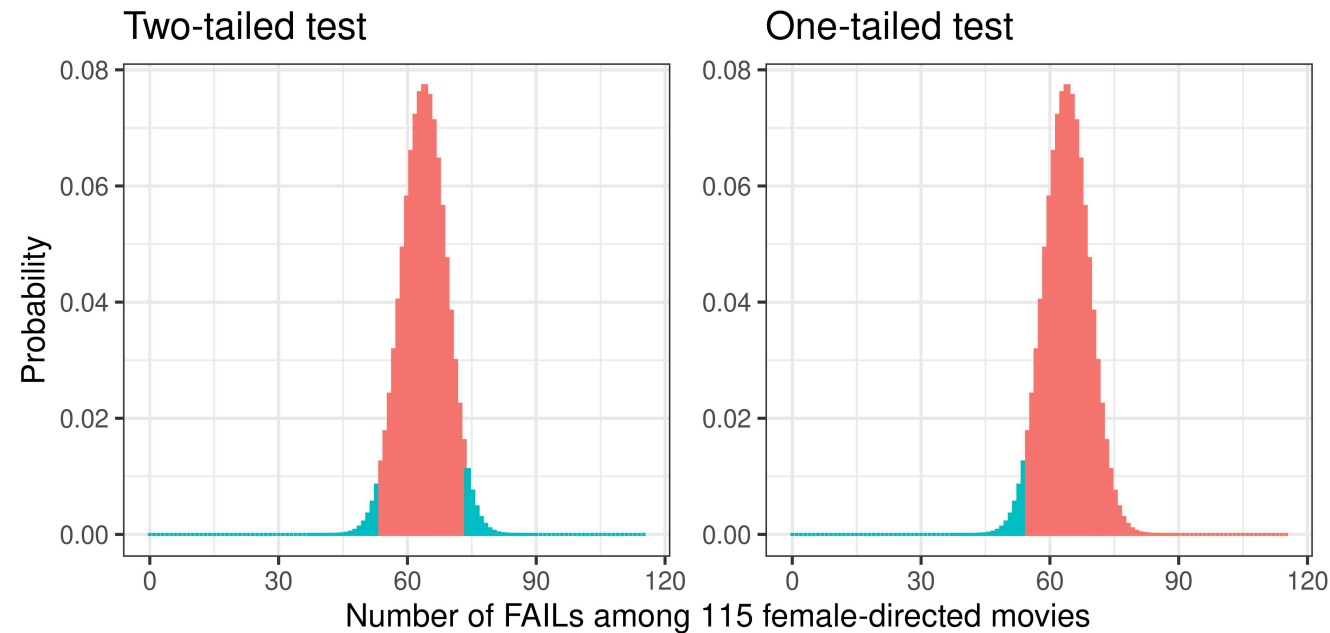
$\mathcal{H}_{1a}$ is an *undirected* hypothesis, while $\mathcal{H}_{1b}$ is directed.

This matters for the question we ask from the sampling distribution!



One tailed vs. Two tailed testing

The alternative hypothesis determines the tail area of the sampling distribution.



For many symmetric continuous tests, the two-tailed p -value is twice the corresponding one-tailed p -value.



Sampling Distributions

Previous weeks, we observed:

$$x_1, x_2, \dots, x_n$$

$$x_{n+1}, x_{n+2}, \dots, x_{2n}$$

$$x_{2n+1}, x_{2n+2}, \dots, x_{3n}$$

etc.

And eventually the empirical distribution of these x approaches the population distribution, say $\mathcal{N}(\mu, \sigma^2)$.



Sampling Distributions

Now we examine:

$x_1, x_2, \dots, x_n$, compute $\bar{x}_1$

$x_{n+1}, x_{n+2}, \dots, x_{2n}$, compute $\bar{x}_2$

$x_{2n+1}, x_{2n+2}, \dots, x_{3n}$, compute $\bar{x}_3$

repeat and for each sample i , compute the sample mean $\bar{x}_i$.

What is the distribution of $\bar{x}_1, \bar{x}_2, \dots$?

- Each sample mean consists of *finite* observations
- The sampling distribution describes how these behave if we'd repeat this experiment infinitely many times
- We look where our sample statistic falls in this distribution



Sampling Distributions

In general, the *sampling distribution* is:

The distribution of a statistic across hypothetical repeated samples from the same population.

Examples of statistics include:

- a proportion
- a correlation
- a sample mean, $\bar{x}$
- a difference between means

For NHST, we care about the sampling distribution *assuming $\mathcal{H}_0$ is true*.



Sampling Distributions

For our Bechdel example, we could derive the sampling distribution exactly.

But suppose instead we observe

$$x_1, x_2, \dots, x_n$$

and are interested in the sample mean

$$\bar{x}.$$

What does the sampling distribution of $\bar{x}$ look like?

Deriving the exact sampling distribution for every possible population distribution would be difficult.



Central Limit Theorem



Central Limit Theorem

Suppose $x_1, x_2, \dots, x_n \stackrel{\text{iid}}{\sim} \mathcal{D}(\mu, \sigma^2, \dots)$.

Then, as n becomes large,

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \approx \mathcal{N} \left(\mu, \frac{\sigma^2}{n} \right).$$

This holds for all population distributions $\mathcal{D}$, as long as they have a finite mean μ and variance σ^2 .

Definitions:

- $\stackrel{\text{iid}}{\sim}$ means *independent and identically distributed*.
- *independent*: “knowing the outcome or value of one variable or event provides no information about the probability or outcome of another.”, i.e., observing x_2 has no influence on what x_7 .
- *identically distributed*: All the x_i come from the same population distribution



Central Limit Theorem

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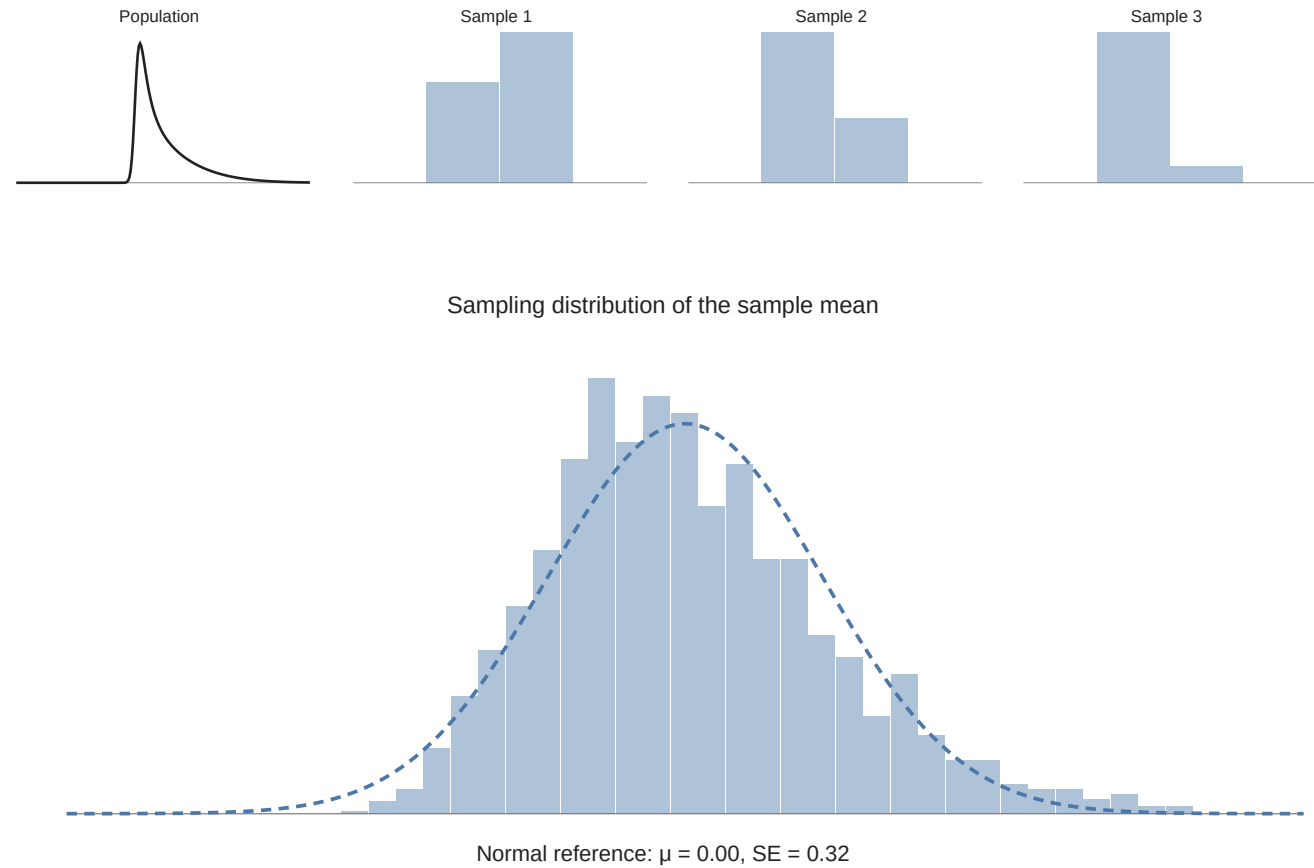
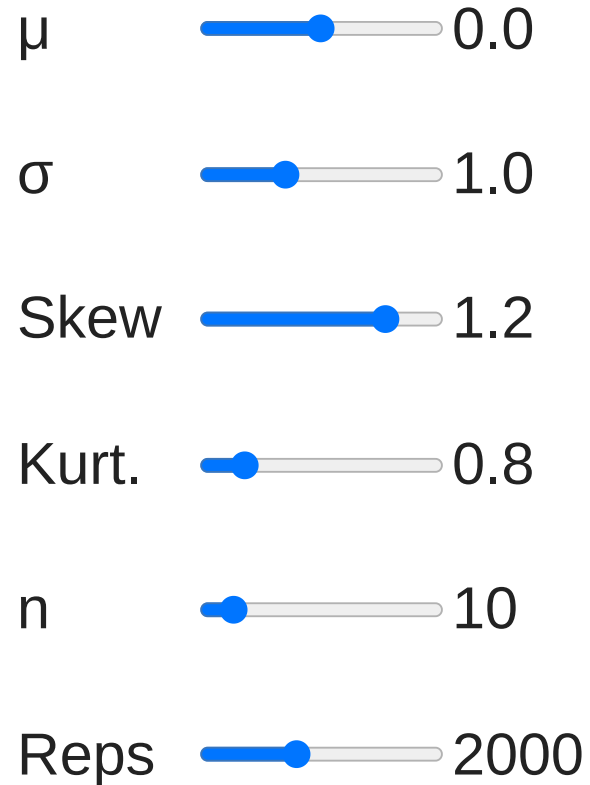
Here, $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$ is known as the *standard error*.

Note that $\sigma_{\bar{x}}$ goes to 0 as n grows.

This indicates that the sample mean converges to the population mean as the no. samples increases.



The sampling distribution of the mean





Why the Central Limit Theorem matters

The impact of $\bar{x} \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$ is huge!

Sums and averages are all around us. For example:

- In educational science, we study the performance of pupils.
 - Even though grades are scored 1 - 10, we examine the sum, so CLT applies
- fMRI machines' raw data are voxels (3D pixels) of the brain.
 - Preprocessing steps involve averaging and smoothing the data across multiple voxels. CLT applies.
- We often study proportions, such as the proportion of movies that pass the Bechdel test.
 - Each response can be coded as $x_i = 0$ or $x_i = 1$.
 - The proportion is the average: $\hat{p} = \frac{1}{n} \sum_i x_i$.
 - Therefore, for sufficiently large n , $\hat{p}$ is approximately normally distributed.



Central Limit Theorem – Addendum

Sometimes, we can derive the sampling distribution *exactly*.

Assume $x_1, x_2, \dots, x_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$.

- Then $\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$.
- But as n tends to ∞ , t_{n-1} tends to $\mathcal{N}(0, 1)$.

So the CLT still applies, but when available we use more precise results for the sampling distribution.

Examples are the t -distribution, F -distribution, χ^2 -distribution.

Some of these we'll revisit in the second half of the course.



Sampling Distribution: Summary

Recall NHST:

1. Define $\mathcal{H}_0$ and pick α .
2. Assume $\mathcal{H}_0$ is true.
3. Ask: how surprising are our observed data under $\mathcal{H}_0$?
4. $p = P(\text{data at least this extreme} \mid \mathcal{H}_0)$
5. If $p < \alpha$, reject $\mathcal{H}_0$.
6. The sampling distribution tells us how unusual our observed statistic is under $\mathcal{H}_0$
7. The Central Limit Theorem gives a very practical approximation to the sampling distribution



Statistical Power



Statistical Power

α is the Type-I error rate: The probability of rejecting a true $\mathcal{H}_0$.

β is the Type-II error rate: The probability of failing to reject a false $\mathcal{H}_0$.

$1 - \beta$ is the power: The probability of rejecting a false $\mathcal{H}_0$.



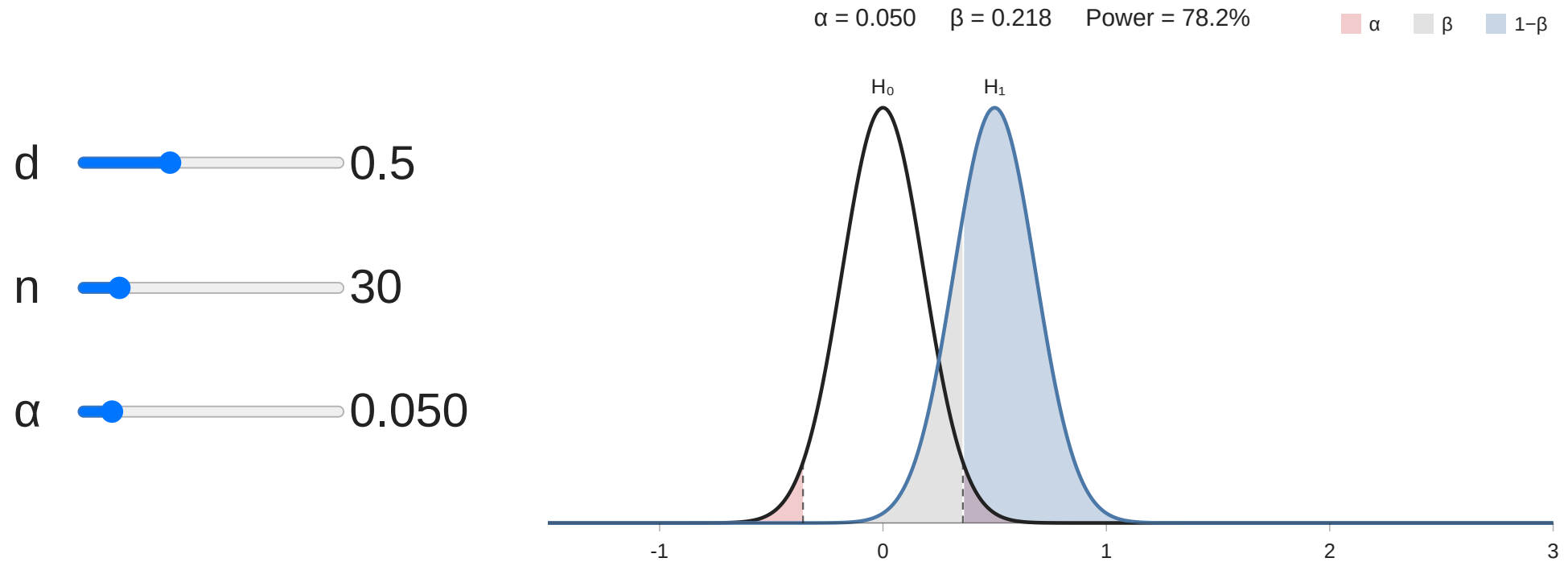
Statistical Power

H_0 FALSE		H_0 TRUE
Reject H_0	Correct rejection / power $1 - \beta$	Type I error α
Do not reject H_0	Type II error β	Correct non-rejection



Statistical Power: Demo

Sampling distributions under $\mathcal{H}_0$ and $\mathcal{H}_1$





Statistical Power: Summary

- α is the Type-I error rate: The probability of rejecting a true $\mathcal{H}_0$.
- β is the Type-II error rate: The probability of failing to reject a false $\mathcal{H}_0$.
- power ($1 - \beta$) is the probability of rejecting a false $\mathcal{H}_0$.
 - Larger effects are easier to detect.
 - A larger sample size increases the power to detect a given effect.
 - A larger α increases the power, but also increase the type-I error rate.



Addendum: Contingency test

We only count FAILs among female-directed movies. Why?

If we keep the row and column totals fixed, we have only one unknown:

DIRECTOR GENDER			
Test	Female	Male	Total row
FAIL	X	$978 - X$	978
PASS	$115 - X$	$783 - (115 - X)$	783
Total column	115	1646	1761

If $X = 21$, 21 female-directed movies FAIL, then $978 - 21 = 957$ male-directed movies must FAIL.